

ROBOTICS AND AI

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Class X Assignment 1-4

Due Date: 20.06.26

40 marks

INSTRUCTIONS

- *Don't copy from AI apps*
 - *Remaining topics will be covered during the summer break.*
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Assignment 1 New Age Robotic Systems

Question A1: Who coined the term 'Robot'?

(1 mark)

Note: Know about contributions of all of them. Then choose the right option

- | | |
|-----------------|---------------------------|
| (a) Asimov | (b) John McCarthy |
| (c) Josef Capek | (d) Joseph F. Engelberger |

Question A2: What is the primary goal of the industrial robots?

(1 mark)

Note: Choose the right option.

- | | |
|------------------------|----------------------------|
| (a) To decrease costs | (b) To increase production |
| (c) To increase safety | (d) All of these |

Question A3: Smart class room is a teaching space which has:

- (i) Smart monitor with a touch control panel system
- (ii) Projector and screen
- (iii) PC/Laptop and internet connection
- (iv) Specialized software

(1 mark)

Select the correct answer from bellow

- | | |
|-----------------------------|----------------------------|
| (a) (i),(ii),(iii) and (iv) | (b) (i),(ii) and (iv) only |
| (c) (iii) and (iv) only | (d) (i) and (ii) only |

Question A4: World's first robot citizen is _____. (1 mark)

Note: Choose the right option.

- | | |
|------------|-------------|
| (a) Apollo | (b) Prepper |
| (c) Tiago | (d) Sophia |

Question A5: Name the Robot from its use: (1 mark)

Note: Write the generic name or an example

- (a) Assist humans in space.
- (b) The designs of the robots are inspired from the nature and biology.
- (c) A robot that helps us in household works.
- (d) A robot that helps to pick objects and to keep it at a particular place.

Question A6: Robots are becoming smarter day by day. Identify 3 unique new age smart robots (NARS) and write their benefits and applications. (3 mark)

Question A7: Make a table on the difference between a machine/robot/cobot. (2 marks)

Assignment 2

From Robots to Cobots

Question B1: In "Cobot", "Co" represents: (1 mark)

Note: Choose the right option.

- | | |
|-------------------|------------------|
| (a) Constructive | (b) Coordinative |
| (c) Collaborative | (d) Controlled |

Question B2: Which one of the following is a limitation of cobots? (1 mark)

Note: Choose the right option.

- (a) Safety approval of cobots
- (b) Supports reduced energy consumption
- (c) Handle heavy loads
- (d) High speed

Question B3: The collaborative robot arms are designed to mimic the range of motion of a _____ (1 marks)

Note: Choose the right option.

- | | |
|-----------------|---------------|
| (a) machine arm | (b) networks |
| (c) device | (d) human arm |

Question B4: What are the distinguishing characteristics of a cobot? (2 mark)

Question B5: Write the importance and limitations of cobots. (5 mark)

Assignment 3

Components of Robots as a System: Gears

Question C1: What is the gear ratio of a gear train with a 20-tooth driving gear and a 40-tooth driven gear? (1 mark)

Hint: Higher gear ratio used in speed reduction. choose the right answer from bellow.

- | | |
|---------|---------|
| (a) 2:1 | (b) 1:2 |
| (c) 1:1 | (d) 3:1 |

Question C2: Gear ratio of a gear train is 3:1. How do you explain the speed transmission and torque amplification? The driven gear will have: (1 mark)

Note: Choose the right option.

- (a) greater speed and greater torque
- (b) lesser speed and greater torque
- (c) greater speed and lesser torque
- (d) none of these above

Question C3: If a 30-tooth driver gear rotates at 300 RPM, how many teeth does the driven gear have if it rotates at 200 RPM? (3 mark)

- | | |
|--------|--------|
| (a) 45 | (b) 30 |
| (c) 20 | (d) 15 |

Question C4: Write the principles of gears. (2 marks)

Question C5: What is a gear? Explain pitch circle, pressure angle. (3 marks)

Assignment 4

Python: Recap

Question D1: Python Programming: Create a list of 10 students in your class and sort them in alphabetical order. (3 marks)

Question D2: Write a program that takes a string as input and returns the count of vowels (a, e, i, o, u) in the string. (3 marks)

Question D3: Write a program that takes a string as input and checks if it's a palindrome (a word that reads the same backwards and forward). Ignore spaces and cases. (4 marks)